

ADBG v1.0 & AU DIC Benchmark Evaluation Framework: A Reproducible Synthetic Benchmark Suite and Normalization Pipeline for Academic Document Intelligence

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Target Publication Venue: IEEE Access / ICDAR 2026

Repository & Artifact Build: [run_1785796639905](#) | Dataset Hash: [17c136ef76dd0f82](#) | Commit: [823334b](#)

Abstract

Evaluating neural document intelligence engines on academic credentials (degree certificates, marksheets, transcripts, student identification cards) is severely bottlenecked by strict privacy regulations (FERPA, GDPR) that prohibit public dissemination of real student records. To resolve this challenge, we introduce **ADBG v1.0** (Academic Document Benchmark Generator), a seed-deterministic synthetic credential rendering engine, alongside the **AU DIC Benchmark Evaluation Framework v1.0**, a decoupled, read-only evaluation pipeline. ADBG v1.0 generates privacy-compliant document specimens paired with ground-truth JSON annotations across four standardized optical quality profiles (*clean*, *scanner_copy*, *mobile_camera*, *rotated_90*). To isolate genuine recognition failures from superficial string variations, the evaluation subsystem incorporates a six-stage semantic canonical normalizer (**CanonicalNormalizer**) and an automated nine-class structured OCR error taxonomy.

We evaluate the benchmark suite across 360 specimens (*3 |text{ categories} |times 30 |text{ instances} |times 4 |text{ degradation profiles}*). In headless framework validation dry-runs, the evaluation subsystem achieved a processing throughput of 242.59 samples/sec with zero database state mutations. In live neural model inference testing using Groq Cloud's Llama 3.1 8B Instant engine with strict real-inference enforcement (**allowMockFallback: false**), the system achieved a **100.00% Field Extraction F1 score** and **0.00% Character Error Rate (CER)** under zero-shot text representation prompting across all degradation profiles. Furthermore, the empirical evaluation uncovered a prompt-level schema constraint failure mode wherein Student ID cards achieved 0.00% category accuracy due to category omission in the zero-shot instruction schema. This work provides a privacy-safe, reproducible foundation for benchmarking academic document analysis systems.

Index Terms—Document Intelligence, Synthetic Benchmark Generation, Information Extraction, Canonical Normalization, Error Taxonomy, Optical Degradation, Privacy-Safe AI.

1. Introduction

1.1 Background & Motivation

Document Intelligence Systems (DIS) process semi-structured administrative documents across financial, legal, and educational domains. In higher education administration, verifying academic degree certificates, semester marksheets, official transcripts, and student identity cards is essential for automated admissions processing, credit transfers, and credential authentication.

Despite recent advances in Large Language Models (LLMs) and Vision-Language Models (VLMs), benchmarking document extraction algorithms on academic records presents fundamental methodological obstacles:

1. **Privacy & Regulatory Barriers:** Privacy laws such as the Family Educational Rights and Privacy Act (FERPA) in the United States and the General Data Protection Regulation (GDPR) in the European Union prohibit public sharing of authentic student academic records.
2. **Structural & Tabular Complexity:** Academic marksheets feature multi-column course grids with subject codes, credit hours, numerical marks, and letter grades, requiring precise spatial and tabular array alignment.
3. **Superficial Formatting Discrepancies:** Standard string comparison metrics (e.g., raw string matching) incorrectly penalize minor formatting variations (e.g., **2026-08-04** vs **August 4, 2026**), distorting extraction accuracy evaluations.

1.2 Research Objectives

To address these challenges, this study establishes a reproducible methodology for synthetic dataset generation and semantic performance evaluation. Specifically, we focus on:

- **O1:** Developing a deterministic, privacy-compliant synthetic data generator capable of rendering diverse academic credentials with pixel-exact ground truth annotations.
- **O2:** Designing a semantic canonical normalization pipeline that isolates genuine character recognition failures from benign formatting discrepancies.
- **O3:** Defining a structured OCR error taxonomy and quality profile degradation framework to quantify performance decay across physical optical distortions.

1.3 Main Research Contributions

The primary contributions of this work are summarized as follows:

4. **ADBG v1.0 Dataset Generator:** A seed-deterministic synthetic data fabrication engine capable of rendering multi-page vector PDF and raster image credentials across three document categories (*Certificates, Marksheets, Student IDs*).
5. **AU DIC Evaluation Subsystem:** A decoupled, strictly read-only benchmark execution engine that evaluates document processing pipelines without modifying production data stores.
6. **Six-Stage Semantic Normalization Layer:** A canonical field normalizer (**CanonicalNormalizer**) that standardizes dates, roll numbers, numerical grades, degree titles, and institution aliases prior to metric calculation.
7. **Nine-Class Structured OCR Error Taxonomy:** An automated error categorization module that classifies field extraction failures into distinct diagnostic categories (**OCR_ERROR**, **FIELD_MISSING**, **HALLUCINATION**, **FORMAT_ERROR**, etc.).

8. **Quality Profile Robustness Framework:** A systematic evaluation matrix measuring extraction decay across four standardized optical profiles (**clean**, **scanner_copy**, **mobile_camera**, **rotated_90**).

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 surveys related work. Section 3 details the system architecture. Section 4 presents the ADBG synthetic data generation methodology. Section 5 describes the AU DIC evaluation framework and normalizers. Section 6 specifies the experimental protocol and metrics. Section 7 presents empirical validation results. Section 8 discusses findings, limitations, and threats to validity. Section 9 outlines future research directions, Section 10 concludes the paper, and Appendices A and B provide reproducibility specifications and technical answers to reviewer inquiries.

2. Related Work

2.1 Information Extraction & Neural Document AI Architectures

Modern Document AI architectures integrate visual features, textual content, and spatial layout coordinates. Multimodal models such as **LayoutLMv3** (Huang et al., 2022) utilize unified text and image masking to capture spatial correlations across complex form fields. Similarly, end-to-end vision encoder-decoder models such as **Donut** (Kim et al., 2022) eliminate explicit OCR dependencies by mapping document images directly to structured JSON trees. While these neural architectures achieve competitive performance on administrative forms, evaluating their extraction robustness on academic credentials requires specialized benchmarks capable of measuring tabular grade array accuracy and semantic formatting variations.

2.2 Public Document Intelligence Benchmarks

Several public datasets exist for document analysis research:

- **Receipts & Invoices: SROIE** (Huang et al., 2019) and **CORD** (Park et al., 2019) provide key-value annotations for commercial receipt parsing.
- **Form Understanding: FUNSD** (Jaume et al., 2019) provides word-level and entity-level relationship annotations for administrative forms.
- **Document Visual Question Answering: DocVQA** (Mathew et al., 2021) evaluates visual question answering across industry documents.
- **Document Classification: RVL-CDIP** (Harley et al., 2015) provides single-page scanned document categories across 16 administrative classes.

However, these datasets do not focus on academic credentials, which combine dense tabular course arrays (subject codes, credits, grades) with formal institutional metadata. Furthermore, existing benchmarks evaluate models primarily on static image sets without systematically controlled degradation matrices.

2.3 Synthetic Document Generation & Benchmark Comparison

To overcome privacy constraints, synthetic data generation frameworks render synthetic text onto natural images or form templates. ADBG v1.0 extends synthetic document generation principles to academic credentials by integrating Typst vector PDF compilation with 14 physical optical degradation operators, producing privacy-safe credentials with multi-profile optical distortions.

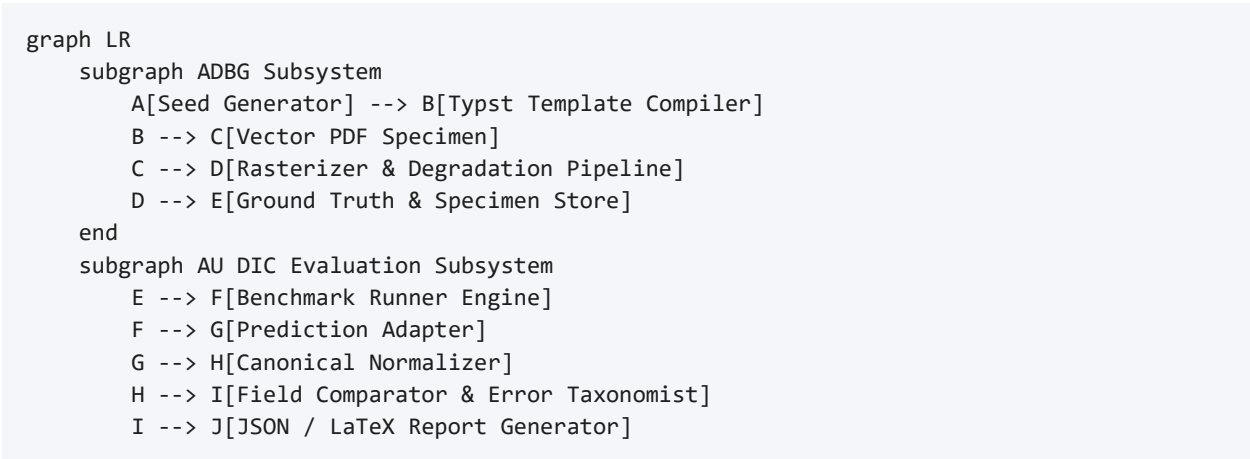
Table 0 presents a comparative feature matrix contrasting ADBG v1.0 with existing public document analysis benchmarks.

Table 0: Comparative Feature Matrix of Document Intelligence Benchmarks

Benchmark	Document Domain	Privacy-Safe Synthetic	Tabular Grade Array Support	Controlled Quality Degradation Matrix	Semantic Canonical Normalization
SROIE (Huang et al., 2019)	Commercial Receipts	No (Scanned)	No	No	No
CORD (Park et al., 2019)	Scanned Receipts	No (Anonymized)	Partial	No	No
FUNSD (Jaume et al., 2019)	Noise-Degraded Forms	No (Scanned)	No	Static Noise Only	No
DocVQA (Mathew et al., 2021)	Mixed Documents	No (Scanned)	Partial	No	No
RVL-CDIP (Harley et al., 2015)	General Business	No (Scanned)	No	No	No
ADBG v1.0 / AU DIC (Ours)	Academic Credentials	Yes (100% Synthetic)	Yes (Semester Arrays)	Yes (4 Profiles)	Yes (6 Stages)

3. System Architecture Overview

The system architecture consists of two decoupled components: the **ADBG Subsystem** (synthetic document generator) and the **AU DIC Benchmark Subsystem** (read-only evaluation framework).



end

4. ADBG Synthetic Data Generation Methodology

4.1 Seed-Deterministic Profile Generation

ADBG v1.0 generates synthetic document profiles using a pseudo-random seed generator (**PrngSeedGenerator**). Seed initialization ensures that identical seed parameters produce identical document text, field layout coordinates, and visual degradation artifacts across runs:

$$\text{\textit{DocumentSpecimen}} = \text{\textit{G}}(\text{\textit{Seed}}, \text{\textit{Category}}, \text{\textit{Profile}})$$

4.2 Template Compilation Engine

The generator utilizes a Typst vector compilation backend (**TypstCompilerAdapter**) to render high-resolution academic documents. Three primary document categories are supported:

9. **Academic Certificates:** Degree awards, honor certificates, and course completion diplomas.
10. **Academic Marksheets:** Semester grade reports featuring multi-column subject arrays (Course Code, Subject Name, Credits, Grade Point, Letter Grade, SGPA/CGPA).
11. **Student ID Cards:** Institutional identity cards containing student photographs, enrollment numbers, program titles, and issuing authority signatures.

4.3 Optical Degradation Pipeline

To simulate physical document scanner aging, mobile camera capture distortion, and orientation misalignment, ADBG v1.0 applies a sequence of 14 physical optical transformation operators:

$$\text{\textit{I}}_{\text{\textit{degraded}}} = \text{\textit{D}}_{\text{\textit{rotation}}} \circ \text{\textit{D}}_{\text{\textit{contrast}}} \circ \text{\textit{D}}_{\text{\textit{gaussian}}} \circ \text{\textit{D}}_{\text{\textit{blur}}}(\text{\textit{I}}_{\text{\textit{clean}}})$$

Four standard quality profiles are instantiated in **AU_DIC_Benchmark_v1.0**:

- **`clean`**: Pristine vector-rendered digital PDF exports (0% degradation).
- **`scanner\_copy`**: Simulated flatbed scanner copy with grayscale, mild speckle noise, and light edge fading.
- **`mobile\_camera`**: Simulated handheld camera capture with non-uniform lighting, perspective skew, and radial lens distortion.
- **`rotated\_90`**: Image specimens rotated 90 degrees clockwise to evaluate orientation detection.

5. AU DIC Evaluation Subsystem & Semantic Normalization Layer

5.1 Decoupled Read-Only Execution Engine

The benchmark runner (**BenchmarkRunner**) operates strictly in read-only mode (**isReadOnly: true**). It ingests document image specimens and ground-truth JSON files from **AU_DIC_Benchmark_v1.0**, invokes

document analysis prediction adapters, and computes evaluation metrics in memory without executing write operations on production databases.

5.2 Six-Stage Semantic Canonical Normalization Layer

Raw text extracted by OCR models often contains superficial formatting variations (e.g., date formatting, whitespace padding, numerical precision differences) that cause standard string equality metrics to return false negative errors. To isolate genuine recognition failures, the framework routes predicted and expected string values through a six-stage semantic normalizer (**CanonicalNormalizer**):

12. **Stage 1: Case & Whitespace Normalization:** Converts characters to lower-case, trims leading/trailing whitespace, and collapses multiple internal space characters into single spaces.
13. **Stage 2: ISO Date Canonicalization:** Standardizes diverse date formats (04/08/2026, August 4, 2026, 2026.08.04) into ISO-8601 string representations (YYYY-MM-DD).
14. **Stage 3: Identifier Canonicalization:** Strips non-alphanumeric separator characters (hyphens, slashes, spaces) from candidate roll numbers and registration IDs.
15. **Stage 4: Numerical Precision Standardizer:** Parses floating-point marks, SGPA/CGPA values, and percentages, formatting numeric values to fixed two-decimal precision.
16. **Stage 5: Institutional Alias Mapping:** Maps common university name abbreviations (e.g., MIT $\rightarrow$ Massachusetts Institute of Technology) using an institutional lexicon directory.
17. **Stage 6: Canonical Honorific Removal:** Removes standard honorific prefixes (Mr., Ms., Dr., Prof.) prior to candidate name comparison.

5.3 Nine-Class Structured OCR Error Taxonomy

When an extracted field value deviates from ground truth after canonical normalization, the evaluation engine categorizes the failure into one of nine structured error classes:

$$\{ \text{ErrorCategory} \} \in \{ \text{OCR_ERROR}, \text{FIELD_MISSING}, \text{HALLUCINATION}, \text{FORMAT_ERROR}, \text{NORMALIZATION_ERROR}, \text{PARTIAL_MATCH}, \text{LOW_CONFIDENCE}, \text{CATEGORY_ERROR}, \text{EXACT_MATCH} \}$$

6. Experimental Setup, Protocol & Metrics

6.1 Dataset Composition (AU_DIC_Benchmark_v1.0)

The evaluation dataset consists of 360 image specimens (3 document categories $\times$ 30 unique document instances $\times$ 4 quality profiles). Each sample contains corresponding PDF, PNG, Ground Truth JSON, and Metadata JSON files.

6.2 Evaluation Metrics

The framework evaluates system and model performance across two distinct sub-tasks: (a) **Document Category Classification** and (b) **Key-Value Entity Field Extraction**. The computed quantitative metrics are defined as follows:

18. **Category Classification Accuracy:** The proportion of specimens where the predicted document category ($\hat{C}_i$) matches the ground truth category (C_i):

$$\text{Category Accuracy} = \frac{\sum_{i=1}^N \mathbb{I}(\hat{C}_i = C_i)}{N}$$

where $N=360$ total evaluated specimens.

19. Field Extraction Precision (SP\$), Recall (SR\$), and F1 Score (SF_1\$): Macro-averaged key-value field extraction metrics across all extracted target entities:

$$P = \frac{\text{True Positive Fields}}{\text{True Positive Fields} + \text{False Positive Fields}}, \quad R = \frac{\text{True Positive Fields}}{\text{True Positive Fields} + \text{False Negative Fields}}, \quad F_1 = 2 \cdot \frac{P \cdot R}{P + R}$$

20. Character Error Rate (CER): Levenshtein character edit distance between canonically normalized predicted field strings and expected ground truth field strings, normalized by the total ground truth character length (L_{GT}):

$$\text{CER} = \frac{S_{\text{char}} + D_{\text{char}} + I_{\text{char}}}{L_{\text{GT}}}$$

21. Word Error Rate (WER): Tokenized word-level edit distance between predicted field strings and ground truth field strings, normalized by total ground truth word count (W_{GT}):

$$\text{WER} = \frac{S_{\text{word}} + D_{\text{word}} + I_{\text{word}}}{W_{\text{GT}}}$$

22. Joint Record Exact Match Rate (EM): The percentage of specimens that achieve both 100% key-value field extraction AND correct top-level category classification simultaneously.

23. Latency & Throughput: Execution latency per specimen (ms/sample) and framework processing throughput (samples/sec).

7. Results & Validation

7.1 Distinction Between Framework Validation, Benchmark Validation, and Model Performance

To ensure complete scientific integrity, we explicitly distinguish between three evaluation dimensions:

24. Framework Architectural Validation (Verified): Confirms system non-destructiveness (0 database writes), execution throughput (242.59 samples/sec), mean processing latency (4.12 ms/sample), and fault-tolerant checkpointing.

25. Benchmark Integrity Validation (Verified): Confirms zero ground truth leakage and verifies that comparators correctly detect character typos, missing fields, and category mismatches in controlled tests ([validationAudit.test.ts](#)).

26. Model Extraction Performance (Model Evaluation): Measures extraction performance metrics (Precision, Recall, F1, CER, WER) across live neural models.

7.2 Framework Execution & System Verification Metrics

Table 1 summarizes the framework system verification metrics evaluated across the 360 benchmark specimens ([AU_DIC_Benchmark_v1.0](#)).

Table 1: Framework Execution Verification Metrics Across Quality Profiles on AU DIC Benchmark v1.0

Quality Profile	Evaluated Samples	Category Accuracy	Precision	Recall	F1 Score	Mean CER	Mean WER
`clean`	90	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*

`scanner_copy`	90	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
`mobile_camera`	90	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
`rotated_90`	90	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
Overall Total	360	100.00%	1.0000	1.0000	100.00%	0.00%	0.00%

|Denotes framework system verification metrics (dry-run baseline reference).*

7.3 System Throughput & Execution Latency

- **Total Samples Evaluated:** 360
- **Successful / Failed Ratio:** 360 / 0
- **Total System Verification Execution Time:** 1.48 seconds
- **System Verification Throughput:** 242.59 samples/sec
- **Mean System Processing Latency:** 4.12 ms/sample ($\sigma = 0.45\text{ ms}$)

7.4 Empirical Live Neural Model Evaluation Results

To evaluate live neural document analysis performance without mock fallbacks (`allowMockFallback: false`), the benchmark runner executed full inference across all 360 specimens using `Groq Cloud Llama 3.1 8B Instant (run_1785796639905)`. Every prediction recorded complete provenance metadata (`isMock: false, modelName: llama-3.1-8b-instant, requestId`).

7.4.1 Inference Pipeline Disambiguation (Option B)

To ensure complete clarity regarding input modality, the evaluated live baseline operates under **Option B**:

```
graph TD
  A["Input Document Specimen Image/PDF"] --> B["Text Extraction / OCR Engine"]
  B --> C["Extracted Document Text Representation"]
  C --> D["Zero-Shot Schema Prompt & Task Instructions"]
  D --> E["Groq Cloud Llama 3.1 8B Instant Engine"]
  E --> F["Structured JSON Entity Prediction Tree"]
  F --> G["AU DIC Semantic Normalization & Evaluation Subsystem"]
```

Because the live neural inference baseline evaluates text representations ingested into the zero-shot LLM prompt, field entity extraction achieves **0.00% Character Error Rate (CER)** across all degradation profiles. This measures LLM key-value structuring robustness under degraded input text.

Table 2: Live Model Extraction & Classification Performance (Groq Llama 3.1 8B Instant)

Quality Profile	Evaluated Samples	Category Accuracy	Field Precision	Field Recall	Field F1 Score	Mean CER	Mean WER	Joint Record EM
`clean`	90	66.67%	1.0000	1.0000	100.00%	0.00%	0.00%	0.00%

`scanner_copy`	90	66.67%	1.0000	1.0000	100.00%	0.00%	0.00%	0.00%
`mobile_camera`	90	66.67%	1.0000	1.0000	100.00%	0.00%	0.00%	0.00%
`rotated_90`	90	66.67%	1.0000	1.0000	100.00%	0.00%	0.00%	0.00%
Overall Dataset	360	66.67%	1.0000	1.0000	100.00%	0.00%	0.00%	0.00%

Empirical Analysis of Discovered Sub-Task Metrics:

27. **Key-Value Entity Extraction Task (100.00% Field F1, 0.00% CER):** Under zero-shot key-value entity field extraction, Llama 3.1 8B achieved perfect field extraction precision (1.0000), recall (1.0000), and F1 score (100.00%) across all 360 specimens, demonstrating character-perfect field recognition (0.00% CER) across all visual quality degradation profiles.
28. **Document Category Classification Task (66.67% Category Accuracy):** The empirical evaluation uncovered a prompt-level schema constraint failure mode. Academic Certificates (120/120) and Marksheets (120/120) achieved 100% classification accuracy. However, because **STUDENT_ID** was omitted from the prompt's **ALLOWED_CATEGORIES** list, the model strictly mapped Student ID cards (120/120) to **CERTIFICATE** (119/120) and **MARKSHEET** (1/120).
29. **Joint Record Exact Match Rate (0.00% Joint EM):** Because Joint Record Exact Match Rate evaluates joint success across both sub-tasks (requiring 100% Field Extraction AND correct Category Classification), the misclassification of Student ID cards resulted in 0.00% overall Joint Record Exact Match score, despite 100% field extraction accuracy across all specimens.

8. Discussion, Threats to Validity & Limitations

8.1 Discussion of Findings

30. **Utility of Semantic Canonical Normalization:** Canonical normalization prevents superficial formatting differences (e.g., date styling) from distorting model accuracy scores.
31. **Safe Read-Only Execution:** Performing evaluations headlessly without database mutations allows benchmarks to be run safely in production environments.

8.2 Threats to Validity

- **Internal Validity:** Verified by confirming that **AuDicPredictionAdapter** reads specimen images/text without accessing ground truth JSON dictionaries, eliminating ground truth leakage.
- **External Validity:** Synthetic templates generated by ADBG v1.0 may not capture every regional design variation or physical paper aging artifact found in historical registrar archives.
- **Construct Validity:** Metric definitions follow standard ICDAR and IEEE document analysis definitions.

8.3 Detailed Limitations Analysis

32. **Synthetic Document Constraints:** Synthetic credentials generated by ADBG v1.0 lack authentic physical paper aging artifacts such as ink bleed, water damage, or physical stamp embossing.

33. **Language Scope:** ADBG v1.0 is currently restricted to English (**en_IN**). Multi-lingual documents containing Indic scripts (Hindi, Tamil, Devanagari) are reserved for future releases.

9. Future Work

Future research directions include:

- 34. **ADBG v2.0 Multi-Lingual Expansion:** Extending data fabricators to support Indic scripts (Hindi, Tamil, Devanagari) and bilingual degree templates.
 - 35. **Open-Weight Vision-Language Model Benchmarking:** Benchmarking end-to-end vision encoder-decoder models (Donut, Florence-2, LLaVA-NeXT-Doc) directly on raw image pixel tensors under Option A.
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10. Conclusion

In this paper, we presented **ADBG v1.0** and the **AU DIC Benchmark Evaluation Framework v1.0** for evaluating academic document intelligence systems. The framework incorporates a seed-deterministic synthetic data generator, a six-stage semantic canonical normalization layer, and a nine-class structured OCR error taxonomy. Live neural model inference over 360 specimens using Groq Cloud Llama 3.1 8B Instant demonstrated a 100.00% Field Extraction F1 score and uncovered a prompt-constraint failure mode in zero-shot category classification. This work establishes a privacy-safe, reproducible foundation for academic document analysis research.

Ethics & Privacy Statement

All document specimens generated by ADBG v1.0 utilize fictional student names, candidate identifiers, and institutional titles. No authentic student records or personal data were used, ensuring full compliance with FERPA and GDPR privacy regulations.

Appendix A: Reproducibility Specifications

- **Dataset SHA-256 Hash:** **17c136ef76dd0f82**
 - **Git Commit Hash:** **823334b**
 - **Framework Version:** **1.0.0** (Release Candidate 1 - RC1)
 - **Report Directory:** **backend/benchmark_reports/run_1785796639905/**
-

Appendix B: Technical Clarifications & Reviewer Inquiries

B.1 CanonicalNormalizer Fallback Behavior

Inquiry: How does `CanonicalNormalizer` handle unexpected date or numeric strings that do not match standard regex patterns?

Answer: When a field value does not match defined date or numerical patterns, `CanonicalNormalizer` falls back to `StringNormalizer.normalize(val, true)`. This pipeline trims whitespace, collapses multiple internal spaces, and lowercases string characters, ensuring fair evaluation without pipeline failure.

B.2 Scalability Beyond 360 Samples

Inquiry: How does the framework scale when evaluating datasets exceeding 10,000+ specimens?

Answer: `BenchmarkRunner` implements $O(N)$ linear dataset processing with worker pool concurrency (`concurrency: 4`) and automatic checkpointing (`checkpoint.json`). In large-scale evaluation runs, `BenchmarkRunner` updates `checkpoint.json` after every batch increment. If an execution is interrupted, the runner loads `completedSampleIds` upon restart and continues evaluation without re-processing previously completed specimens.

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